

Python & SQLite Databases

Learn how to store and manage data with Python!

A fun coding lesson for young coders



1. Why Do We Need Databases?

Databases help us keep lots of information safe and organized — like a digital filing cabinet!

- Store thousands of records neatly
- Find information quickly
- Keep data safe even when the app closes

2. Meet SQLite!

SQLite is a tiny database that lives right inside your Python programs!

- No setup needed — just import and go!
- Stores everything in one small file
- Perfect for learning and small projects

3. How Does Python Talk to SQLite?

Python uses special commands to work with SQLite databases.

- We can **Add** new data
- We can **Read** existing data
- We can **Update** or **Delete** data too!

4. Setting Up SQLite in Python

Just 3 simple steps to get started!

- Step 1: `import sqlite3`
- Step 2: Connect to a database file
- Step 3: Create a cursor to run commands

5. Creating a Database & Table

A table is like a spreadsheet inside your database.

- `CREATE TABLE` makes a new table
- We add columns like name, age, class
- Each row is one record (like one student)

6. Adding Data (Insert)

Use `INSERT` to add new records to your table!

```
cursor.execute("""
    INSERT INTO students
    (name, age, class)
    VALUES ("Ananya", 10, "5A")
""")
```

This adds a new student named Ananya to our database!

7. Reading Data (Select)

Use `SELECT` to read and display data from your table!

- `SELECT * FROM students` shows all records
- You can also pick specific columns
- Results come back as a list

```
cursor.execute("SELECT * FROM students")
results = cursor.fetchall()
for row in results:
    print(row)
```

8. Updating Data

Use `UPDATE` to change existing records in your table!

- `UPDATE` changes data that already exists
- Use `WHERE` to pick which record to change
- Always save changes with `commit()`

```
cursor.execute("""
    UPDATE students
    SET age = 11
    WHERE name = "Ananya"
""")
```

9. Deleting Data

Use `DELETE` to remove records you no longer need.

- `DELETE FROM` removes a record
- Use `WHERE` to pick which one to delete
- Be careful — deleted data is gone!

```
cursor.execute("""
    DELETE FROM students
    WHERE name = "Ananya"
""")
```

10. Where is Python + SQLite Used?

- Mobile apps that save your progress
- School systems that store grades
- Games that remember your score
- Websites that keep your login info

11. What Next? Practice Ideas

- Make a database of your favourite books
- Store your friends' birthdays
- Create a quiz score tracker
- Build a mini library app!

Quiz Time!

20 Multiple Choice Questions — Circle the correct answer!

Q1.

What is a database used for?

- A.** Playing music
- B.** Storing and organizing information
- C.** Drawing pictures
- D.** Sending emails

Q2.

SQLite is best described as:

- A.** A programming language
- B.** A tiny database inside your Python program
- C.** A type of computer
- D.** A web browser

Q3.

Which Python module do you import to use SQLite?

- A.** sqlmodule
- B.** database3
- C.** sqlite3
- D.** pymysql

Q4.

What does SQLite store all its data in?

- A.** Many separate folders
- B.** One small file
- C.** The internet
- D.** A printed book

Q5.

Which command creates a new table?

- A.** MAKE TABLE
- B.** NEW TABLE
- C.** CREATE TABLE
- D.** BUILD TABLE

Q6.

In a database table, each row represents:

- A.** A column name
- B.** One single record
- C.** The whole database
- D.** A picture

Q7.

Which command is used to add new data to a table?

- A.** ADD
- B.** INSERT
- C.** PUT
- D.** APPEND

Q8.

Which command is used to read data from a table?

- A.** READ
- B.** SHOW
- C.** SELECT
- D.** GET

Q9.

What does `SELECT * FROM students` do?

- A.** Deletes all students
- B.** Shows all records in the students table
- C.** Creates a new student
- D.** Renames the table

Q10.

Which command changes existing data in a table?

- A.** CHANGE
- B.** UPDATE
- C.** FIX
- D.** EDIT

Q11.

Which keyword helps you pick which record to update or delete?

- A.** WHERE
- B.** WHEN
- C.** WHICH
- D.** PICK

Q12.

Which command removes a record from a table?

- A.** REMOVE
- B.** CLEAR
- C.** DELETE
- D.** ERASE

Q13.

What should you always do after making changes to save them?

- A.** print()
- B.** save()
- C.** commit()
- D.** close()

Q14.

In Python, what do we use to run SQL commands?

- A.** A cursor
- B.** A loop
- C.** A list
- D.** A print statement

Q15.

What does `cursor.fetchall()` return?

- A.** A single number
- B.** A list of all matching results
- C.** An error message
- D.** A new table

Q16.

True or False: SQLite is big and hard to use.

- A.** True
- B.** False
- C.** Only on weekends
- D.** Only for adults

Q17.

Which of these is NOT something a table column might store?

- A.** name
- B.** age
- C.** class
- D.** a live video

Q18.

Which real-world example uses Python + SQLite?

- A.** A game that remembers your score
- B.** A paper notebook
- C.** A pencil
- D.** A wall clock

Q19.

What is the first step to set up SQLite in Python?

- A.** Create a cursor
- B.** import sqlite3
- C.** Delete a table
- D.** Print the results

Q20.

Why is learning Python and databases a superpower?

- A.** It helps you build games and apps that save information
- B.** It makes your computer fly
- C.** It replaces your teacher
- D.** It only works on Mondays

✓ Answer Key

1. **B**

2. **B**

3. **C**

4. **B**

5. **C**

6. **B**

7. **B**

8. **C**

9. **B**

10. **B**

11. **A**

12. **C**

13. **C**

14. **A**

15. **B**

16. **B**

17. **D**

18. **A**

19. **B**

20. **A**

🚀 **Thank You!** 🚀

Thanks for learning Python & SQLite with us!
Any questions?

Keep Coding! 🚀